

Ky Fan Tail Majorization and a Weaker Endpoint Corridor to Stage A

Captured Energy, Rank Staircases, and Seven-Scale Stress Evidence

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Abstract

RH-83 proved that termwise singular-value majorization is sufficient for an optimal endpoint factorization, but dyadic singular staircases make that condition stronger than the downstream problem requires. This paper isolates the weaker invariant that RH-78 actually uses: the Hilbert–Schmidt energy left outside a low-rank postblock space.

For a Hilbert–Schmidt operator B with Gramian $G = B^*B$, Ky Fan’s principle gives

$$\tau_r(B)^2 = \operatorname{tr} G - \sum_{j=1}^r \lambda_j(G) \leq \|B(I - P)\|_2^2$$

for every rank- r orthogonal projector P . Thus a validated captured-energy lower bound certifies the optimal tail without resolving each singular value.

If the physical postblock state satisfies the tail-majorization condition

$$\tau_{J_\sigma + \ell}(B_\sigma) \leq \alpha_\sigma \tau_{J_\sigma + \ell}(\mathcal{R}_\sigma) + \varepsilon_\sigma,$$

then RH-82 immediately yields an exponential excess-rank tail. Polylogarithmic α_σ , remainder, and observability bounds close the RH-78 effective-rank corridor. This condition neither identifies coordinates nor controls the leading singular values, and is strictly weaker than the RH-83 factorization gate.

The numerical audit keeps the five 192-bit interval-certified postblock tails and adds two out-of-sample levels: $\sigma = 0.005$ at dimension 1024 and $\sigma = 0.0025$ at dimension 2048. The clock rank grows only from four to eight. At every level the physical clock-plus-two tail is below 1.4% of the linear endpoint-model tail. The largest interval-certified relative tail is 2.34×10^{-7} ; the two extended binary64 stress tails are below 1.56×10^{-15} .

The result narrows the all-level problem to a captured-energy theorem for a clock-dimensional postcritical packet space. The two new levels are diagnostic, not a proof of uniformity. Stage A1, unconditional Stage A4, Hilbert–Polya, and the Riemann Hypothesis remain open.

Keywords: Ky Fan principle; singular-value tail; effective rank; captured energy; postblock state; small noise.

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1 Why termwise majorization is more than needed

The RH-83 optimal factorization theorem compares every leading physical singular value with the corresponding endpoint singular value [4]. This is natural for an operator factorization, but individual ratios can jump when a new singular staircase enters between neighboring dyadic scales. RH-78 does not use those ratios. It uses only the residual left after a low-rank approximation [2].

RH-82 already supplies the mediator estimate

$$\tau_{J_\sigma + \ell}(\mathcal{R}_\sigma) \leq C_R q^\ell, \quad J_\sigma = \frac{\log(1/\sigma)}{2 \log \lambda} + O(1), \quad (1)$$

with $0 < q < 1$ [3]. The question is how little information about B_σ is required to inherit this tail.

2 Ky Fan certification of a postblock tail

Let $B : \mathcal{H}_1 \rightarrow \mathcal{H}_2$ be Hilbert–Schmidt and let $G = B^* B$. Define

$$\tau_r(B) = \inf_{\text{rank } L \leq r} \|B - L\|_2.$$

Theorem 2.1 (Ky Fan captured-energy upper). *For every integer $r \geq 0$,*

$$\tau_r(B)^2 = \text{tr } G - \sum_{j=1}^r \lambda_j(G). \quad (2)$$

If P is any rank- r orthogonal projection on $\mathcal{H}_1$, then

$$\boxed{\tau_r(B)^2 \leq \|B(I - P)\|_2^2 = \text{tr } G - \text{tr}(PG)}. \quad (3)$$

Proof. The first identity is the Eckart–Young formula. Ky Fan’s maximum principle states

$$\text{tr}(PG) \leq \sum_{j=1}^r \lambda_j(G).$$

Subtract from $\text{tr } G$ to obtain (3) [1]. □

The theorem is suited to interval work: one may fix a floating candidate subspace and evaluate the residual directly in rigorous arithmetic. No interval eigendecomposition of the tiny tail eigenvalues is required.

3 Tail-majorization transfer

Theorem 3.1 (Weak endpoint tail corridor). *Suppose that for a rank schedule $r_\sigma = J_\sigma + \ell_\sigma$,*

$$\tau_{r_\sigma}(B_\sigma) \leq \alpha_\sigma \tau_{r_\sigma}(\mathcal{R}_\sigma) + \varepsilon_\sigma. \quad (4)$$

Then

$$\boxed{\tau_{r_\sigma}(B_\sigma) \leq \alpha_\sigma C_R q^{\ell_\sigma} + \varepsilon_\sigma}. \quad (5)$$

If α_σ , ε_σ , and the square root of the future observability norm are polylogarithmic, then a logarithmic or polylogarithmic r_σ satisfies the RH-78 effective-rank residual gate.

Proof. Insert (1) into (4). Multiplication by the future observability factor transfers the Hilbert–Schmidt residual to the directional Hardy future exactly as in RH-77. □

This is strictly weaker than an RH-83 factorization. A factorization implies (4) by the ideal property, but a tail inequality says nothing about the leading singular vectors or even their individual sizes.

4 Seven-scale audit

The first five levels retain the 192-bit Arb residuals from RH-82. The audit then builds two new frozen matrices at constant spatial resolution per noise width:

$$(\sigma, n, M) = (0.005, 1024, 49), \quad (0.0025, 2048, 64).$$

These horizons continue the proposed log-square schedule. They are not an analytic horizon theorem.

For each channel, the endpoint model uses the linear powered-row Gram matrix and the physical rank is

$$r_\sigma = \lceil H_\sigma \rceil + 2. \quad (6)$$

σ	fine dimension	rank	max physical/endpoint tail	evidence
0.16	32	4	1.393×10^{-2}	Arb
0.08	64	5	3.814×10^{-6}	Arb
0.04	128	6	9.969×10^{-7}	Arb
0.02	256	6	5.143×10^{-6}	Arb
0.01	512	7	9.257×10^{-7}	Arb
0.005	1024	8	4.687×10^{-12}	binary64 stress
0.0025	2048	8	1.050×10^{-12}	binary64 stress

Table 1: Maximum over both directional channels. Only the first five physical tails are interval certified.

The worst ratio occurs at the coarsest scale. At smaller noise the physical tail is much smaller than the already small endpoint tail. The ambient dimension grows by a factor 64, while the tested rank only doubles.

5 Next theorem and boundary

The all-level target can now be stated without singular-value matching: construct a rank- $O(\log(1/\sigma))$ postcritical packet projector P_σ and prove

$$\text{tr}(P_\sigma B_\sigma^* B_\sigma) \geq \|B_\sigma\|_2^2 - \text{polylog}(1/\sigma)^2. \quad (7)$$

Combined with Theorem 2.1, this is enough. The packet vectors may be rotated by the dynamics; they need not equal sampled endpoint rows.

This paper proves the Ky Fan and tail-transfer theorems and validates the five inherited interval scales plus two floating stress levels. It does not prove (7) uniformly, close Stage A1 or unconditional Stage A4, construct an A5 relative determinant, produce a self-adjoint Hilbert–Polya operator, derive a $T \log T$ or prime-power law, identify zeta zeros, or prove the Riemann Hypothesis.

References

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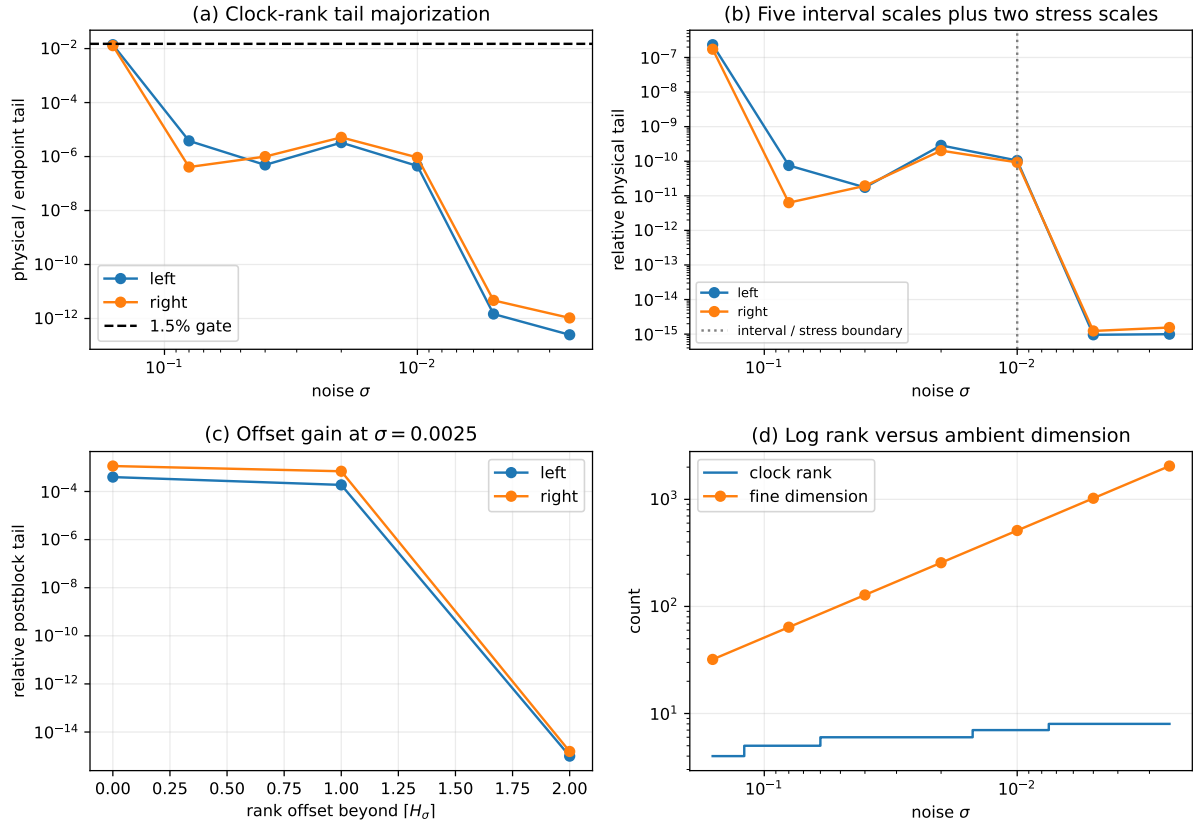


Figure 1: Tail-majorization ratios, interval and stress residuals, excess-rank gain at the finest level, and logarithmic rank versus ambient dimension.